

ALLAM NEHAL VARMA

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Education

Vellore Institute of Technology (VIT), Vellore

B.Tech in Computer Science and Engineering

Sep 2022 – May 2026

GPA: 7.89/10.0

Bharathi Junior College

Intermediate (MPC) | Telangana State Board

Aug 2020 – May 2022

Score: 90.1%

Experience

SmartInternz (in collaboration with Google Developers)

Jun 2024 – Jul 2024

Machine Learning Intern

Remote

- Delivered an end-to-end **TensorFlow** pipeline; ran **30+** experiments and improved validation accuracy from **0.78** → **0.86** via tuning (learning rate, batch size, regularization).
- Automated preprocessing (imputation, scaling, augmentation), cutting data prep time by **45%** and eliminating **12** manual steps with reproducible scripts.
- Packaged a **SavedModel** and CLI inferencer; median CPU inference latency **90 ms**; shipped a **20-page** doc with metrics and deployment notes (**score: 88/100**).

VIT Vellore University

Jan 2024 – Jan 2025

Research Intern (under Prof. Sivakumar V.)

Vellore, India

- Conducted research on anomaly detection in IoT systems using **GRU**-based time-series models.
- Authored and published “*GRU in Anomaly Detection for IoT: A Comparative Study*” in **IEEE SCEECs’25**, presented at MANIT Bhopal.
- Explored evaluation methodologies and comparative analyses for sequence modeling and anomaly detection.

Academic Projects

Concrete Strength Prediction API

May 2024 – Jul 2024

- Tech:** Python, Flask, Pandas, NumPy, scikit-learn, Matplotlib
- Trained on **1,030** records with **8** inputs; engineered **4** derived features and applied **5-fold CV** over **40+** configs; selected model with $R^2 = 0.906$, **RMSE = 89** (**22%** RMSE drop vs. baseline).
- Exposed **5** REST endpoints (`/predict`, `/health`, etc.); median inference **< 250 ms**; added **12+** unit tests and **joblib** serialization for fast cold starts.

Code Summarization Tool

Sep 2024 – Dec 2024

- Tech:** LLM (OpenAI), FastAPI, Python, ReactJS, MongoDB
- Created** a tool that generates **concise, relevant summaries** from large inputs (code, research papers, API docs) to surface key insights quickly.
- Engineered an **iterative prompting** pipeline that improved contextual accuracy over zero-shot and simple prompt-chaining approaches.
- Designed a React frontend and FastAPI backend with MongoDB persistence for requests, results, and metadata.

AI Image Classification Service with SDN-based Load Balancing

Jul 2025 – Sep 2025

- Tech:** Python (FastAPI), Docker/Containernet, **SDN** (Ryu/OpenFlow), MobileNetV2
- Scaled inference from **1** to **4** replicas; throughput increased from **26 req/s** to **85 req/s**.
- SDN load balancing reduced **p95 latency** from **520 ms** to **230 ms** and cut **HTTP 5xx** errors by **90%**.

Technical Skills

Technical Skills: Java, C, Python, JavaScript, ReactJS, MongoDB, Selenium, PL/SQL, MySQL, R, C++, Machine Learning, Spring Boot.

Others: Git, Linux, Tableau, AWS, Docker, RStudio, MATLAB, Red Hat Linux 9, LLM, Flask, WordPress, IoT, REST API